Key Generation Ceremony Documentation Package

Index & Overview

This document serves as the authoritative index and overview for the Key Generation Ceremony (KGC) Documentation Package, outlining the complete set of materials, procedures, and evidentiary records associated with the secure generation of cryptographic key material. It is designed to ensure full traceability, transparency, and auditability of all activities performed before, during, and after the ceremony.

The package consolidates engagement documentation, procedural controls, hardware procurement records, operational runbooks, witness statements, attestation reports, and post-event audit logs into a single reference framework. Its structure adheres to best practices defined by ISO/IEC 27001, and NIST SP 800-57, providing a standardized foundation for cryptographic key lifecycle governance. Additional reference frameworks include the The Capital Markets and Technology Association (CMTA) Digital Assets Custody Standard (DACS) and the CryptoCurrency Security Standard (CCSS).

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Distribution List	[Client] Management Team

Purpose_

This document provides a structured overview and reference index for all materials generated as part of the Key Generation Ceremony (KGC) for [Client]. It ensures compliance with internal security policies, ISO/IEC 27001, and NIST SP 800-57 standards, and facilitates auditability and reproducibility of the ceremony process.

Scope of the Key Generation Ceremony_

This Key Generation Ceremony (KGC) governs the controlled creation and management of multiple BIP39 seed phrases, each independently generated using verifiable entropy sources. These seed phrases are subsequently XOR-combined to derive a single master seed[1], which serves as the root of trust for the client's cryptographic wallet infrastructure.

The resulting key material is based on Elliptic Curve Digital Signature Algorithm (ECDSA) constructs and will form the foundation for deterministic wallet derivations and signing operations. The master key will further function as a Bring Your Own Key (BYOK) input for integration into other secure systems, such as Hardware Security Modules (HSMs), Multi-Party Computation (MPC) environments, or custodial vault frameworks, ensuring consistency and traceability across cryptographic domains.

All processes within this scope are executed under strict procedural, physical, and logical controls to ensure non-repudiation, entropy integrity, and recoverability, with complete documentation for future key backup, rotation, and attestation events.

[1] By using multiple seeds and xor'ing their value, we avoid a Single Point of Failure (SPOF), meaning no single individual within the organization would have single access to the cryptographic material at any point in time.

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Version Control and Change Management_

Version	Date	Author	Description
0.1.0	2025-10-22	José J. Perez Aguinaga	Initial release.
	Date		
	Date		
	Date		

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Document Package Structure_

#	Name	ID	Type	Function	Status	Remarks
1	**Engagement Letter**	XXX-KGC-2025-002	Contractual	Legal	Draft	Defines scope and liabilities.
2	**Key Ceremony Policy**	XXX-KGC-2025-003	Policy	Security	Not issued	Defines procedural requirements.
3	**Runbook (Procedure)**	XXX-KGC-2025-004	Procedure	Ops/Sec	Draft	Steps to conduct KGC.
4	**Hardware Procurement List**	XXX-KGC-2025-005	Record	Procurement	Draft	Vendor and device details.
5	**Environment Setup Checklist**	XXX-KGC-2025-006	Checklist	Security	Draft	Verifies controlled setup.
6	**Participant NDAs and Roles**	XXX-KGC-2025-007	Contractual	Compliance	Not issued	Confidentiality and duties.
7	**Attestation Report (AUP)**	XXX-KGC-2025-008	Report	Auditor	Not issued	Auditor's verifications.
8	**Witnesses Declarations**	XXX-KGC-2025-009	Record	Security	Not issued	Signed witnessed declarations.
9	**Chain of Custody Form**	XXX-KGC-2025-010	Record	Custody	Not issued	Storage media tracking.
10	**Audit Trail Log**	XXX-KGC-2025-010	Log	Ops/Sec	Not issued	Chronological log of events.
11	**Integrity Manifest**	XXX-KGC-2025-011	Annex	Security	Not issued	Visual and hash integrity proof.
12	**Summary Report**	XXX-KGC-2025-012	Report	Auditor	Not issued	Consolidated overview.

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References_

List of reference standards or frameworks used:

Frameworks

- ISO/IEC 27001:2022 - Information Security Management Systems (ISMS)
- ISO/IEC 11770-1:2010 - Key Management Techniques
- NIST SP 800-57 Part 1 Rev. 5 - Key Management Guidelines

Standards

- CCSS v9.0 - Cryptocurrency Security Standard
- CMTA's Digital Assets Custody Standard (DACS)

Traceability Matrix_

Requirement	Standard Reference	Corresponding Document
Ceremony Conduct	ISO/IEC 27001 A.12.1.1	Runbook, AUP
Cryptographic Key Management	NIST SP 800-57 §5.1	Runbook, Policy
Personnel Confidentiality	ISO/IEC 27001 A.7.1.2	NDA
Asset Management	ISO/IEC 27001 A.8.1.1	Procurement List, Chain of Custody
Audit and Accountability	ISO/IEC 27001 A.12.4	Logs, Witness Statements

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